THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** puberty	Puberty marks the beginning of reproductive maturity.
2	**(d)** lower temperature	Heat flows downhill in temperature.
3	**(b)** second	1 minute = 60 seconds.
4	**(a)** ingestion	Ingestion is intake.
5	**(a)** puberty	Reproductive maturity.
6	**(c)** pass on heat to neighbours but stay in position	They vibrate.
7	**(a)** distance travelled per unit time	Speed = distance/time.
8	**(a)** villi	They increase surface area.
9	**(a)** growth spurt	Sudden increase.
10	**(a)** more	They absorb more radiation.
11	**(a)** odometer	Shows kilometres covered.
12	**(d)** breathing	Intake of O2 and release of CO2.
13	**(d)** 11 to 19 years	Teenage years.
14	**(d)** conduction, convection, radiation	Standard modes.
15	**(c)** 3600 seconds	60 x 60.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Period when body becomes sexually mature (1); About 10 to 12 years (1).	1+1
17	Conduction and convection (1); Radiation (1).	1+1
18	Distance travelled per unit time (1); m/s (1).	1+1
19	Ingestion, digestion, absorption (1); Assimilation and egestion (1).	1+1
20	Monthly discharge of blood from the uterus (1); It occurs if the egg is not fertilised (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Rapid increase in height (1); Voice change (1); Growth of body hair (1).	1+1+1
22	Fix pins with wax on a rod (1); Heat one end (1); Pins fall one by one (1).	1+1+1
23	T = 60/30 = 2 s (1); Frequency = 30/60 (1); 0.5 per second (1).	1+1+1
24	Mouth to oesophagus (1); Stomach to small intestine (1); Large intestine to anus (1).	1+1+1
	OR: Secretes adrenaline (1); Helps in emergencies (1); Increases heartbeat (1).	1+1+1
25	Hot air rises (1); Cool air sinks (1); Convection spreads it (1).	1+1+1
	OR: 72 x 5/18 = 20 m/s (1); 15 x 18/5 (1); 54 km/h (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Growth spurt (1); Voice change in boys (1); Facial hair in boys (1); Breast development in girls (1); Menstruation in girls (1).	1+1+1+1+1
	OR: Conduction: spoon in tea (1); Convection: boiling water (1); Radiation: Sun (1); Conduction needs solid (1); Radiation needs no medium (1).	1+1+1+1+1
27	Hang pendulums of lengths 25 cm and 100 cm (1); Time 20 oscillations each (1); Find T = t/20 (1); Longer has greater period (1); Mass has no effect (1).	1+1+1+1+1

	OR: Mouth: chewing, saliva (1); Oesophagus: carries food (1); Stomach: digestion of proteins (1); Small intestine: digestion and absorption (1); Large intestine: water absorption (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** gills	Dissolved oxygen.
2	**(a)** voice box (larynx)	Larynx enlarges.
3	**(a)** moving particles of the fluid	Moving fluid.
4	**(d)** metre per second	m/s.
5	**(b)** digestion	Digestion.
6	**(a)** in general girls	Girls begin earlier.
7	**(b)** light-coloured clothes	They reflect heat.
8	**(d)** speedometer	Shows km/h.
9	**(a)** liver	It makes bile.
10	**(b)** teenage	Ages 13 to 19 are teen years.
11	**(d)** conduction	Heat passes from particle to particle.
12	**(d)** time period	Time for one to-and-fro swing.
13	**(a)** nostrils	Nose.
14	**(d)** Adam's apple	A lump in the throat.
15	**(b)** radiation	Waves can travel in vacuum.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	They reflect more heat (1); Keep us cool (1).	1+1
17	400/80 (1); 5 m/s (1).	1+1
18	Finger-like projections in the small intestine (1); Increase surface for absorption (1).	1+1
19	Sweat glands are more active (1); To avoid infections and body odour (1).	1+1
20	Metals conduct heat (1); Food cooks faster (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Heart contracts and relaxes (1); Blood pushed into arteries (1); Veins return blood (1).	1+1+1
22	Healthy bones and muscles (1); Good mood (1); Controls weight (1).	1+1+1
23	Trap air (1); Insulation (1); Retain body heat (1).	1+1+1
24	20 min = 1/3 h (1); 18 x 1/3 (1); 6 km (1).	1+1+1
	OR: Cockroach: spiracles and tracheae (1); Fish: gills (1); Both take oxygen (1).	1+1+1
25	Master gland (1); Controls other glands (1); Releases growth hormone (1).	1+1+1
	OR: Water at the bottom gets hot (1); Hot water rises (1); Cool water sinks to replace it (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Growth hormone (1); Thyroxine: goitre (1); Insulin: diabetes (1); Imbalance causes problems (1); Medical help (1).	1+1+1+1+1
	OR: Medium needed (1); Direction (1); Particle movement (1); Example (1); Speed (1).	1+1+1+1+1
27	Oscillation: one to-and-fro (1); Time period: time for one (1); Amplitude: extreme displacement (1); Bob returns (1); Periodic motion (1).	1+1+1+1+1
	OR: Kidneys filter blood (1); Urine in ureters (1); Stored in bladder (1); Passed through urethra (1); Waste removal (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** kilometre per hour	km/h.
2	**(a)** moist skin	Skin.
3	**(d)** bathe and keep clean	Hygiene.
4	**(a)** woollen clothes	Wool traps air.
5	**(d)** distance-time graph	Plots distance vs time.
6	**(b)** absorption	Absorption.
7	**(d)** hormones	Hormones travel in blood.
8	**(b)** convection	Hot fluid rises and cold falls.
9	**(d)** oscillation	Bob returns to the starting point.
10	**(c)** gall bladder	It helps digest fats.
11	**(b)** secondary sexual character	It appears at puberty.
12	**(d)** convection currents	Hot water rises.
13	**(b)** distance / time	Speed.
14	**(c)** lungs	They exchange gases.
15	**(a)** oil glands becoming more active	Hormones increase oil.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Uniform: equal distances in equal time (1); Non-uniform: unequal (1).	1+1
17	Carries oxygen (1); Gives blood red colour (1).	1+1
18	Growth of facial hair (1); Deepening of voice (1).	1+1
19	Transfer of heat through a solid (1); Spoon in hot tea (1).	1+1
20	Time for one complete oscillation (1); Measured in seconds (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	180/4 = 45 km/h (1); 45 x 5/18 (1); 12.5 m/s (1).	1+1+1
22	Teeth chew, saliva acts on starch (1); Stomach churns food (1); Acid and pepsin digest proteins (1).	1+1+1
23	Balanced diet (1); Exercise (1); Adequate sleep (1).	1+1+1
24	Metal conducts heat quickly from hand (1); Wood is a poor conductor (1); We feel the heat loss (1).	1+1+1
	OR: 400/5 = 80 km/h (1); 80 x 3 (1); 240 km (1).	1+1+1
25	Throb of arteries (1); Count at the wrist (1); About 72 per minute (1).	1+1+1
	OR: Body grows during sleep (1); Improves memory (1); Reduces stress (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Pituitary: growth hormone (1); Thyroid: thyroxine (1); Pancreas: insulin (1); Adrenal: adrenaline (1); Testes and ovaries: sex hormones (1).	1+1+1+1+1
	OR: Heat water in a beaker (1); Place crystal at the bottom (1); Coloured streaks rise (1); Move around the beaker (1); Show convection currents (1).	1+1+1+1+1
27	First 60 x 2 = 120 km (1); Second 40 x 3 = 120 km (1); Total 240 km in 5 h (1); Average 240/5 (1); 48 km/h (1).	1+1+1+1+1
	OR: Starch by amylase (1); Proteins by pepsin (1); Fats by lipase and bile (1); Final products absorbed (1); Pancreatic juice helps (1).	1+1+1+1+1

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